

MANGALORE REFINERY AND PETROCHEMICALS LIMITED

ANNUAL STATUTORY PLANT INSPECTION REPORT

Unit: Crude Distillation Unit (CDU-5)

Inspection Scope: Column Feed Piping & Pressure Relief Loops

Date of Inspection: 15-January-2026

Lead Inspector: Senior Inspection Engineer (Badge #MRPL-8821)

EXECUTIVE SUMMARY:

Routine non-destructive testing (NDT) and ultrasonic thickness gauging were executed across Crude Feed Piping Train P-102 and associated bypass manifolds.

CRITICAL FINDINGS:

1. Piping Section Line 04-CR-102 (Pipe P-102): Measured minimum wall thickness of 3.8 mm against nominal 5.0 mm. Baseline 3.5 years ago was 5.0 mm. Minimum retirement thickness threshold per design is 3.0 mm.
2. Isolation Valve V-14: Slight hydrocarbon weeping detected at packing gland seal.
3. Flange Connection F-08: Atmospheric surface corrosion; ultrasonic scan confirms no loss of flange face integrity.

RECOMMENDATION:

Submit technical approval note to Operations In-Charge for scheduled turnaround replacement package.

MRPL MECHANICAL INTEGRITY & NDT FIELD MEASUREMENT LOG

Tag: Line 04-CR-102 (Crude Feed to Column C-101)

Ultrasonic Thickness Measurements (mm):

█ T-Point01 (North Elbow): 4.1 mm

█ T-Point02 (Intrados Throat): 3.8 mm <-- MINIMUM OBSERVED

█ T-Point03 (Straight Run): 4.6 mm

█ T-Point04 (South Reducer): 4.4 mm

Operating Pressure: 14.5 kg/cm2 | Temperature: 285 deg C

Field Inspector Signature: [Verified J. Rao, Level-II UT]